

AS
ECONOMICS
7135/1

Paper 1 The Operation of Markets and Market Failure

Practice Paper — Advanced

Mark scheme

Mark schemes are working documents. Indicative content is provided as a guide; examiners must credit other valid points. Students do not have to cover all of the indicative content to reach the highest level of the mark scheme. An answer which contains nothing of relevance to the question must be awarded no marks.

SECTION A

The following list indicates the correct answers used in marking students' responses.

Q	Ans	Description	Q	Ans	Description
1	B	ration scarce resources between competing uses.	11	B	above the socially optimal level.
2	B	Consumers seek to maximise their utility given their budget.	12	B	sellers know more than buyers about the quality of cars.
3	C	A successful advertising campaign for the good.	13	A	Total revenue rises from £32 to £36.
4	B	Incumbent firms sustaining supernormal profits.	14	B	an excess supply of labour at the minimum wage.
5	A	an increase of approximately 7%.	15	C	the upward-sloping section to the right of the minimum.
6	C	Vaccination reducing the risk of disease for non-vaccinated individuals.	16	D	X and Y are strong substitutes.
7	C	pivots the supply curve, raising it by a larger amount at higher prices.	17	B	Higher labour productivity tends to reduce the unit labour cost of production.
8	B	decrease average total cost.	18	C	a normal luxury good.
9	D	Perfect competition.	19	B	creating a market in the right to pollute and capping total pollution.
10	B	A government subsidy paid per tonne of wheat produced.	20	B	would be under-provided and under-consumed in a free market because consumers undervalue the long-term private and social benefits.

Q5 working: $\% \Delta TR \approx \% \Delta P + \% \Delta Q = +10\% + (-0.3 \times 10\%) = +10\% - 3\% = +7\%$.

Q13 working: At £8, $Q = 4 \rightarrow TR = £32$. At £6, $Q = 6 \rightarrow TR = £36$.

Levels of response — 25 mark questions

Level	Response	Marks
5	Sound, focused analysis and well-supported evaluation that: • is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors • includes good application of relevant economic principles to the given context and, where appropriate, good use of data to support the response • includes well-focused analysis with clear, logical chains of reasoning • includes supported evaluation throughout the response and in a final conclusion.	21–25
4	Sound, focused analysis and some supported evaluation that: • is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors • includes some good application of relevant economic principles to the given context and, where appropriate, some good use of data to support the response • includes some well-focused analysis with clear, logical chains of reasoning • includes some reasonable, supported evaluation.	16–20
3	Some reasonable analysis but generally unsupported evaluation that: • focuses on issues that are relevant to the question, showing satisfactory knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles to the given context and, where appropriate, some use of data to support the response • includes some reasonable analysis but which might not be adequately developed or becomes confused in places • includes fairly superficial evaluation; there is likely to be some attempt to make relevant judgments but these are not well-supported by arguments and/or data.	11–15
2	A fairly weak response with some understanding that: • includes some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • includes some limited application of relevant economic principles to the given context and/or data to the question • includes some limited analysis but it may lack focus and/or become confused • includes attempted evaluation which is weak and unsupported.	6–10
1	A very weak response that: • includes little relevant knowledge and understanding of economic terminology, concepts and principles • includes application to the given context which, at best, is very weak • includes attempted analysis which is weak and unsupported.	1–5

Context 1 — The market for electric vehicles in the UK

2 1 Define 'positive externality'.

[3 marks]

Level	Response	Max 3 marks
3	A full and precise definition is given.	3 marks
2	The substantive content of the definition is correct, but there may be some imprecision or inaccuracy.	2 marks
1	Some fragmented points are made.	1 mark

Examples of acceptable definitions worth 3 marks:

- a benefit from production or consumption that is enjoyed by a third party, not involved in the transaction
- the part of the social benefit not captured by the private decision-maker (where $MSB > MPB$)

Examples of a definition worth 2 marks:

- a benefit to a third party
- an external benefit

Examples of a definition worth 1 mark:

- a benefit not paid for
- an example of a positive externality (e.g. vaccination, education)

MAXIMUM FOR QUESTION 21: 3 MARKS

2 2 Percentage change in BEV registrations between 2020 and 2023.

[4 marks]

Calculation: $[(314.7 - 108.2) \div 108.2] \times 100 = 190.85... = 190.9\%$ (to 1 dp).

Response	Max 4 marks
For the correct answer: 190.9% (or a rise of 190.9%)	4 marks
For the correct value, not rounded to one decimal place: e.g. 190.85% OR for the correct value, rounded the wrong way: 190.8% OR for the correct value but with incorrect units (e.g. £190.9 or 190.9 cars)	3 marks
For the correct calculation/figures throughout but the wrong answer OR an answer with two of: missing/incorrect units, incorrect rounding, missing sign	2 marks
For the correct method (change $\div$ original $\times$ 100) OR a correct intermediate value such as 206.5 (the absolute change in thousands)	1 mark

MAXIMUM FOR QUESTION 22: 4 MARKS

2 3 Two significant features of UK BEV registrations.

[4 marks] Award up to 2 marks for **each** significant feature.

Response	Max 4 marks
Identifies a significant feature Makes accurate use of the data to support the feature identified Unit of measurement given accurately	2 marks

Response	Max 4 marks
Identifies a significant feature but only one piece of data is given when two are needed and/or no unit of measurement is given and/or the unit of measurement is inaccurate and/or the wrong date is given	1 mark

If a candidate identifies more than two features, reward the best two.

Significant features include:

- Registrations are highest in 2023 at 314.7 thousand
- Registrations are lowest in 2018 at 15.5 thousand
- Registrations rise in every year of the period shown
- The range is 299.2 thousand (15.5k to 314.7k)
- The largest absolute increase is between 2020 and 2021, when registrations rose from 108.2k to 190.7k — a rise of 82.5 thousand
- The largest proportional increase is between 2019 and 2020 (from 37.9k to 108.2k — an increase of about 185%)
- Overall, registrations rose from 15.5k in 2018 to 314.7k in 2023, an increase of 299.2 thousand (over 20 times)

Note: Allow a margin of ± 5 thousand.

MAXIMUM FOR QUESTION 23: 4 MARKS

2 4 PPF diagram — productivity increase in EV production only.

[4 marks]

Expected diagram: an initial PPC bowed out from the origin (or a straight line) between two appropriately labelled axes ('Electric vehicles' and 'Other goods and services'). A second PPC should pivot outward along the EV axis only, while still touching the same point on the other goods axis. Curves clearly labelled and/or arrows used to indicate the direction of the shift.

Response	Max 4 marks
Accurately drawn initial PPC and a second PPC pivoting outward along the EV axis only, with appropriately-labelled axes and a clear indication of which curve is which	4 marks
Accurately drawn initial PPC and a pivoted second PPC but no indication of which is which OR PPC with clear indication of pivot but with at least one axis label missing or inappropriate	3 marks
Accurately drawn initial PPC with labelled axes but no, or an incorrect, second curve (e.g. a parallel outward shift)	2 marks
Accurately drawn initial PPC with at least one axis label missing or inappropriate and no, or an incorrect, second curve OR appropriately labelled axes but with incorrect or missing curve(s)	1 mark

Notes: A title is not required. The exact shape does not matter, but if the PPC does not touch both axes deduct 1 mark. The only appropriate axes labels are 'Electric vehicles' and 'Other goods and services'.

MAXIMUM FOR QUESTION 24: 4 MARKS

2 5 Explain how economies of scale could affect the price of electric vehicles in the UK.

[10 marks]

Level	An answer that:	Marks
3	• is well organised and develops one or more of the key issues that are relevant to the question • shows sound knowledge and understanding of relevant economic terminology, concepts and principles • includes good application of relevant economic principles and/or good use of data to support the response • includes well-focused analysis with a clear, logical chain of reasoning • may include a relevant diagram to support their explanation.	8–10
2	• includes one or more issues that are relevant to the question • shows reasonable knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles and/or data to the question • includes some reasonable analysis but it might not be adequately developed and may be confused in places • may include a relevant diagram to support their explanation.	4–7
1	• is very brief and/or lacks coherence • shows some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • demonstrates very limited ability to apply relevant economic principles and/or data to the question • may include some very limited analysis but the analysis lacks focus and/or becomes confused • may include a diagram but the diagram is likely to be inaccurate in some respects or is inappropriate.	1–3

Relevant issues include:

- Definition of economies of scale — the fall in long-run average cost as output expands
- Types of internal economies of scale relevant to EV production — technical (specialised machinery and assembly lines), purchasing (bulk-buying batteries and components), financial (cheaper credit), marketing, managerial
- Significance of falling battery costs — batteries are the most expensive single component of EVs
- External economies of scale — clustering of EV suppliers, government-funded charging infrastructure, training of specialised engineers
- Lower average costs allow firms to lower prices, especially in a competitive market
- Use of a diagram showing the supply curve shifting to the right and a fall in equilibrium price, or a long-run average cost curve falling with rising output
- Recognition that the extent of any price fall depends on the degree of competition, the PED for EVs and the willingness of firms to pass cost savings to consumers
- Possibility of diseconomies of scale at very high volumes

MAXIMUM FOR QUESTION 25: 10 MARKS

2 6 Discuss whether the UK government should provide more support for the EV market.

[25 marks]

Areas for discussion include:

- Recent trends — rapid growth in BEV registrations; supply chain expansion
- EVs as a means of reducing the negative externalities of consumption (air pollution, carbon emissions) from petrol and diesel vehicles
- Diagram of a negative externality of consumption showing the welfare loss from over-consumption of petrol/diesel cars, and how subsidies for EVs (or taxes on ICE cars) can move the market towards the social optimum
- Methods of intervention — purchase grants, VAT relief, subsidised charging infrastructure, tax on petrol/diesel vehicles, congestion charges, scrappage schemes, the ZEV mandate, advertising

regulation

- Effect on different agents — consumers (lower prices, wider choice, income effects), manufacturers (cost of complying with the mandate), workers in the supply chain, low-income households, rural households
- Equity concerns — EVs are still more expensive than petrol cars; high-income households are the main beneficiaries of purchase subsidies (regressive)
- Financial cost of more support to the government, and opportunity cost (NHS, schools, other priorities)
- Risks of government failure — picking winners, dependence on subsidies, manufacturers waiting for higher grants, complexity of the ZEV mandate
- Negative externalities elsewhere in the supply chain — battery mining, end-of-life disposal — that may not be reduced and could be displaced abroad
- Case for leaving more of the transition to the market — falling EV prices via economies of scale, voluntary consumer switching, lower private running costs
- Market failure versus government failure
- An overall, supported judgement

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the candidate's response.

Use the levels mark scheme on the earlier page to award marks for this question.

MAXIMUM FOR QUESTION 26: 25 MARKS

Context 2 — Childcare in the UK

2 7 Define 'variable costs'.

[3 marks]

Level	Response	Max 3 marks
3	A full and precise definition is given.	3 marks
2	The substantive content of the definition is correct, but there may be some imprecision or inaccuracy.	2 marks
1	Some fragmented points are made.	1 mark

Examples of acceptable definitions worth 3 marks:

- costs (expenses) which change with output
- costs incurred when paying for variable factors of production

Examples of a definition worth 2 marks:

- costs (expenses) which change
- total costs minus fixed costs

Examples of a definition worth 1 mark:

- costs (expenses) which are not fixed
- example of a variable cost, e.g. raw materials, electricity

MAXIMUM FOR QUESTION 27: 3 MARKS

2 8 Percentage change in number of childcare providers 2015 → 2023.

[4 marks]

Calculation: $[(68\,100 - 81\,300) \div 81\,300] \times 100 = -16.2361... = -16.2\%$ (to 1 dp).

Response	Max 4 marks
For the correct answer: -16.2% (or a fall of 16.2%)	4 marks
For the correct value without the minus sign: 16.2% OR for the correct value, not rounded to one decimal place: e.g. -16.24% OR for the correct value, rounded the wrong way: -16.3% OR for the correct value but with incorrect units (e.g. $-\pounds 16.2$)	3 marks
For the correct calculation/figures throughout but the wrong answer OR an answer with two of: missing/incorrect units, incorrect rounding, missing sign	2 marks
For the correct method (change $\div$ original $\times 100$) OR a correct intermediate value such as $-13\,200$ (the absolute change)	1 mark

MAXIMUM FOR QUESTION 28: 4 MARKS

2 9 Two significant features of the average weekly cost of full-time nursery childcare.

[4 marks] Award up to 2 marks for **each** significant feature.

Response	Max 4 marks
Identifies a significant feature Makes accurate use of the data to support the feature identified Unit of measurement given accurately	2 marks
Identifies a significant feature but only one piece of data is given when two are needed and/or no unit of measurement is given and/or the unit of measurement is inaccurate and/or the wrong date is given	1 mark

Significant features include:

- Cost is highest in 2023 at £296 per week
- Cost is lowest in 2017 at £232 per week
- The range is £64 per week (£232 to £296)
- Cost rises every year of the period shown
- Cost is higher at the end of the period than at the start, rising from £232 to £296, an increase of £64 per week (about 27.6%)
- The largest single-year increase is between 2022 and 2023, when the cost rose from £279 to £296 — a rise of £17 per week
- The smallest single-year increase is between 2020 and 2021, when the cost rose from £263 to £269 — a rise of £6 per week

Note: Allow a margin of $\pm$ £3.

MAXIMUM FOR QUESTION 29: 4 MARKS**3 0 PPF diagram — trade-off between paid work and unpaid childcare.**

[4 marks]

Expected diagram: an initial PPC bowed out from the origin (or a straight line) between two appropriately-labelled axes ('Paid work' and 'Unpaid childcare'). A second curve outside the first along the paid work axis (or a movement along the original PPC towards more paid work) showing that subsidised childcare allows more paid work for the same amount of unpaid childcare. Allow either interpretation, provided the candidate explains the direction shown.

Response	Max 4 marks
Accurately drawn initial PPC with appropriately-labelled axes and a clearly indicated outward shift along the paid work axis (or a clear movement along the PPC towards more paid work)	4 marks
As above but no indication of direction OR at least one axis label missing or inappropriate	3 marks
Accurately drawn initial PPC with labelled axes but no, or an incorrect, second curve/point	2 marks
Accurately drawn initial PPC with at least one axis label missing or inappropriate and no, or an incorrect, second curve OR labelled axes but missing/incorrect curve(s)	1 mark

Notes: A title is not required. If the PPC does not touch both axes, deduct 1 mark. The only appropriate axis labels are 'Paid work' and 'Unpaid childcare'.

MAXIMUM FOR QUESTION 30: 4 MARKS**3 1 Explain factors that could cause a rise in the price of childcare.**

[10 marks]

Level	An answer that:	Marks
3	• is well organised and develops one or more of the key issues that are relevant to the question • shows sound knowledge and understanding of relevant economic terminology, concepts and principles • includes good application of relevant economic principles and/or good use of data to support the response • includes well-focused analysis with a clear, logical chain of reasoning • may include a relevant diagram to support their explanation.	8–10

Level	An answer that:	Marks
2	• includes one or more issues that are relevant to the question • shows reasonable knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles and/or data to the question • includes some reasonable analysis but it might not be adequately developed and may be confused in places • may include a relevant diagram to support their explanation.	4–7
1	• is very brief and/or lacks coherence • shows some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • demonstrates very limited ability to apply relevant economic principles and/or data to the question • may include some very limited analysis but the analysis lacks focus and/or becomes confused • may include a diagram but the diagram is likely to be inaccurate in some respects or is inappropriate.	1–3

Relevant issues include:

- Recognition that a rise in price can result from a fall in supply and/or a rise in demand
- Rising variable costs (staff wages, food, energy) shifting the short-run supply curve to the left
- Falling number of providers reducing market supply
- Regulatory requirements (staff-to-child ratios, qualifications) raising the cost of provision
- Significance of rising employment of parents and demographic factors raising demand
- Rising real incomes increasing the demand for higher-quality childcare (YED)
- Government subsidies that raise effective demand (the 'free hours' scheme): if providers face capacity constraints, fees for non-subsidised hours may rise
- Use of a relevant supply–demand diagram showing the shift(s) and the resulting rise in equilibrium price
- Recognition of the relatively inelastic demand for childcare which allows providers to pass on cost rises

MAXIMUM FOR QUESTION 31: 10 MARKS

3 2 Discuss whether the UK government should increase its subsidies for childcare.

[25 marks]

Areas for discussion include:

- Early years education as a merit good — long-term private and external benefits
- Positive externalities of consumption — better outcomes for children, higher future productivity, lower future welfare and crime costs
- Diagram of a merit good showing under-consumption in the free market and how subsidies can move output toward the social optimum
- Effects on the labour market — improved labour supply, especially of mothers; effects on the gender pay gap, productivity and economic growth
- Existing schemes (15/30 hours, Tax-Free Childcare) and their limitations — provider complaints that hourly rates are below cost, cross-subsidy from paid hours
- Effects on different agents — parents, children, childcare workers (low pay), providers (especially small private nurseries), taxpayers
- Financial cost of increased subsidies to the government and opportunity cost (NHS, schools, other priorities)
- Equity considerations — whether subsidies should be universal or targeted at low-income families
- Government failure — administrative complexity, providers gaming the system, rising fees for non-subsidised hours, possible drop in quality of provision
- Case for leaving more of the market to operate freely — parental choice, quality competition,

alternatives such as grandparents or childminders

- Equity versus efficiency considerations
- An overall, supported judgement on whether subsidies should be increased and in what form

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the candidate's response.

Use the levels mark scheme on the earlier page to award marks for this question.

MAXIMUM FOR QUESTION 32: 25 MARKS